JOY STUDENT User Guide

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Overview

The Desk Watch JOY STUDENT is a small desk console. Its two-line, sixteen-character screen shows the temperature, the humidity and the time. A joystick on top lets you move from one page to another and set each function.

It has eleven menu entries:

- **Favorite:** the page shown right at start-up.
- **Temp & Hum:** temperature and humidity with alerts.
- **Timer:** a work timer with breaks (Pomodoro method).
- Counter, Stopwatch, Alarm, Dice, Chess clock, Melody.
- Settings and Sleep mode.

The time, the alarm and the automatic night mode depend on the RTC module: without it, these three functions do not work.

Power-up and joystick

Turning the console on

1. Plug the USB cable into a computer, a charger or a power bank.
2. Do not touch the joystick during start-up: the console measures its rest position at that moment.
3. The screen shows “JOY STUDENT”, then the Favorite page.

The console has no switch: to turn it off, unplug the USB cable.

Joystick gestures

Gesture	Effect
Up or down	Scroll the menu, change a value (hold to repeat, faster after 1.5 s)
Right, or press the head	Confirm (OK)
Right or press twice within 0.4 s	Double click
Left	Back (menu, cancel)
Left held 2.5 s	Go to sleep

The double click is mostly used to go back to the menu and to reset to zero. Its exact effect is given page by page below.

The menu

From the Favorite page, move the joystick or press it to open the menu. Move up or down to pick an entry, then confirm with right or a press. The screen changes color depending on the page.

The menu pages

Favorite and Temp & Hum

The page shows the time on the first line, and the temperature and humidity on the second. The background is green when everything is fine and the OK indicator appears on the right.

- When a measurement goes outside its thresholds, a message alternates with the readings: TEMP TOO HIGH or TOO LOW, HUM TOO HIGH or TOO LOW. The background turns red for a high threshold, blue for a low one, and the buzzer sounds.
- **Double click:** sets the four thresholds (low and high temperature from 10 to 40 °C in 0.5 °C steps, low and high humidity from 10 to 90 %). Move up or down to change the value, confirm to go to the next threshold; the last one confirmed saves them.
- If the screen shows “Sensor missing”, check the cable of the SHT sensor.

The Favorite page is the one you choose in the settings: Auto, Temp & Hum, Clock, Timer, Stopwatch or Counter.

Timer

A work timer with automatic breaks (Pomodoro method).

- When stopped, move up or down to set the work time: from 1 to 60 minutes in 1-minute steps, then from 65 minutes to 4 hours in 5-minute steps.
- Confirm to start. Confirm again to pause or resume.
- After each work period comes a 5-minute break, and a 15-minute break every four periods. The buzzer announces each change. The background is red during work, green during a short break, blue during a long break.
- **Double click:** reset to zero; once at zero, a double click goes back to the menu.
- If you open the menu during the countdown, the timer keeps running in the background.

Counter

Each press on the joystick head adds 1, up to 9999. Move the joystick to ask for a reset: up = yes, down = no, or do nothing for 5 seconds to cancel. If the timer or the stopwatch is running, its time is shown at the bottom left.

Stopwatch

Confirm to start, then to stop; confirm again to resume. The time is shown in minutes, seconds and tenths. **Double click:** reset to zero; once at zero, back to the menu. The stopwatch keeps running if you open the menu.

Alarm

The RTC module is required. The screen shows the alarm time and its state, ON or OFF.

- **Double click:** turns the alarm on or off.
- **Confirm:** sets the alarm step by step: hours, minutes, alarm on or off, melody, volume. The melody and the volume are previewed while you choose.
- Eight ringtones are available: Birthday, Fur Elise, Mario Bros, Tetris, Aurora, Cosmic Pulse, Arcade Hero and Epic Rise.
- When the alarm rings, one press snoozes it for 5 minutes. If nobody reacts, the ringtone stops after 2 minutes and is snoozed.
- The alarm rings even when the console is asleep.

Dice

Move up or down to choose the number of sides: 2, 4, 6, 8, 10, 12, 20, 30 or 100. Each press rolls the die; a clicking sound accompanies the roll for about a second, then the result stays on screen. **Double click:** back to the menu.

Chess clock

Choose the time control while stopped (minutes + increment in seconds): 1+0, 3+0, 3+2, 5+0, 5+3, 10+0, 10+5, 15+10, 30+0 or 60+0. Confirm to start; player 1 begins.

- During the game, any movement of the joystick passes the turn to the other player, with a short beep.
- **Double click** (or hold for about 1.2 s): pause. Confirm to resume, double click to reset to zero.
- When a player's time runs out, the screen turns red, shows END and the buzzer sounds three times.
- Leaving the page pauses the game.

Melody

Move up or down to choose a tune (Birthday, Fur Elise, Mario Bros or Tetris), confirm to play, pause or resume. **Double click:** back to the menu. The volume is set in the settings.

Settings and sleep mode

Settings

Move up or down to choose a setting, confirm to change it, confirm again to save it. A double click goes back to the menu and cancels the setting in progress.

Setting	Choices
Brightness	8 levels, from 5 % to 100 %, with an instant preview
Volume	Low, Medium or High
Boot sound	On or off: the start-up melody
Auto sleep	Never, 1, 2, 5, 10, 15, 30 minutes or 1 h
Night mode	No, Auto (from 10 pm to 6 am, with the RTC) or Yes: dimmed screen
Set time	Sets the hours then the minutes of the RTC
Favorite	Auto, Temp & Hum, Clock, Timer, Stopwatch or Counter

All settings are kept when you unplug the console.

Sleep mode

The last entry of the menu, or holding left for 2.5 seconds, turns the screen off after “Goodbye”. With automatic sleep, the console turns itself off after the chosen delay, except during a chess game in progress.

- To wake the console, move the joystick or press it.
- The alarm and the background functions keep running during sleep.
- Temperature and humidity alerts stay silent during sleep.
- Sleep mode turns the screen off, but the console is still powered: unplug the USB cable to stop it completely.

Troubleshooting

Problem	What to do
The screen stays black	Move the joystick: the console may be asleep. Then check the USB cable, the Grove cable of the screen and the brightness in the settings.
The joystick picks things on its own or seems inverted	It was touched at start-up. Unplug the USB cable, plug it back in, and do not touch the joystick until the Favorite page appears.
“Sensor missing”	Check that the SHT sensor is plugged into an I2C socket, then unplug and plug the USB cable back in.
The time shows “-- : --” or the alarm asks for the RTC module	The RTC module is not detected: check its cable on an I2C socket.
No sound	Check the volume and the buzzer on socket D3. Temperature alerts are silent during sleep and in night mode.
The time is wrong	Use Settings, then Set time.
An alert keeps coming back	Set the thresholds with a double click on the Temp & Hum page, so that they suit the room.

If nothing works, note what the screen shows and show it to your teacher.